Analytics Report: Emergency Department Performance

1. Introduction

Emergency departments play a critical role in providing timely medical care. Understanding patient arrivals, waiting times, satisfaction and referral patterns can help hospital administrators assess service performance and identify areas requiring further investigation.

This project analyses emergency department patient records using Microsoft Excel. The objectives are to examine waiting-time patterns, explore the relationship between patient waiting time and satisfaction, and investigate the association between patient race and department referral.

2. Dataset and Data Preparation

The dataset contains 12 variables covering patient demographics, visit information, operational performance and administrative information. The analysis uses 9,216 patient records.

Key variables include Patient Age, Patient Gender, Patient Race, Patient Admission Date, Department Referral, Patient Admission Flag, Patient Satisfaction Score and Patient Waittime. Patient Id was used for record counting and duplicate checks.

The data quality audit found no duplicate patient IDs. Recorded ages ranged from 1 to 79 years, while waiting times ranged from 10 to 60 minutes. The gender categories were F, M and NC; the meaning of NC requires confirmation.

The Patients CM variable was excluded because its meaning could not be established. Patient first initials and last names were not required for the analysis and should not be included in shared outputs.

Patient Satisfaction Score contained 6,699 blank values, representing approximately 72.69% of the 9,216 records. The remaining 2,517 records contained recorded scores ranging from 0 to 10. Blank scores were retained in the source dataset but excluded from satisfaction-specific calculations.

Excel PivotTables, charts and statistical formulas were used to summarise the data and examine the relationships between selected variables.

3. Emergency Department Waiting Times

3.1 Waiting Time by Gender

The average recorded waiting time was 35.11 minutes for female patients, 35.40 minutes for male patients and 37.17 minutes for patients in the NC category.

The difference between the male and female averages was approximately 0.29 minutes. The NC category had the highest recorded average, although its sample size and coding meaning require further investigation.

These are descriptive results and do not establish that gender causes differences in waiting time.

3.2 Waiting Time by Patient Race

Native American/Alaska Native patients had the highest reported average waiting time at 35.69 minutes, while Pacific Islander patients had the lowest at 34.65 minutes. The difference between these averages was approximately 1.04 minutes.

Average waiting times were relatively similar across the seven racial categories. However, similar averages alone do not establish equitable care. Further analysis should consider group sizes, clinical urgency, arrival times and other relevant patient characteristics.

3.3 Patient Arrivals by Hour

The busiest recorded arrival hour was 11:00 p.m. to 11:59 p.m., with 436 patient records, representing approximately 4.73% of the 9,216 records.

This finding identifies a period of comparatively high recorded arrivals. Additional analysis of arrivals and staffing across all hours would be needed before making operational staffing recommendations.

4. Relationship Between Waiting Time and Patient Satisfaction

The correlation coefficient between Patient Waittime and Patient Satisfaction Score was -0.0212.

This indicates a negligible negative linear relationship among the records included in the correlation calculation. In the analysed sample, longer waiting times were not meaningfully associated with lower satisfaction scores in a linear sense.

However, approximately 72.69% of patient records lacked recorded satisfaction scores. This substantial amount of missing information limits the extent to which the result can be generalised to all emergency department patients.

The result does not establish that waiting time has no effect on satisfaction. Other factors, including communication, perceived quality of care, clinical outcomes and expectations, may also influence patient experience. Correlation does not establish causation.

5. Association Between Patient Race and Department Referral

A chi-square test of independence was conducted to examine the association between Patient Race and Department Referral.

The reported results were $\chi^2(42) = 117.49$, $p < .001$. Cramér's V was approximately 0.0461.

The chi-square result indicates a statistically significant association between the two categorical variables. However, Cramér's V indicates that the association is very weak in magnitude.

The referral distribution showed that 58.59% of records were classified as "None." General Practice was the most common named referral category, accounting for 19.97% of records.

The meaning of the "None" category should be verified before concluding that these patients did not require or receive referrals. The observed association also does not establish that race causes differences in referral patterns.

Further analysis should consider clinical needs, patient characteristics, access to care, referral eligibility and other potential explanatory factors.

6. Discussion and Recommendations

The analysis identifies several areas that hospital administrators could investigate further.

First, the reported arrival pattern suggests that late-night demand deserves examination alongside staffing levels and service capacity.

Second, waiting-time averages were relatively similar across the racial categories examined. Further investigation is needed to determine whether differences remain after accounting for clinical urgency, arrival time and other relevant variables.

Third, the negligible correlation between waiting time and satisfaction suggests that waiting time alone does not explain the observed variation in satisfaction scores. However, the high proportion of blank satisfaction scores is an important limitation and should be investigated.

Fourth, the statistically significant but very weak association between patient race and department referral warrants careful interpretation. The hospital could examine referral practices and clinical factors to determine whether meaningful differences in care pathways exist.

Recommended actions include:

- Investigate why satisfaction scores are absent from most records.
- Verify the meanings of the NC gender category and the Patients CM variable.
- Clarify whether “None” means no referral, no referral required or another administrative status.
- Examine patient arrivals and staffing patterns across the full day.
- Consider clinical urgency and other relevant factors when comparing waiting times and referral patterns.
- Continue monitoring patient experience and referral practices using consistent definitions and reliable data.

7. Limitations

The analysis has several limitations. A substantial proportion of satisfaction scores are missing, potentially limiting the representativeness of the correlation result. The meaning of some coded categories remains unconfirmed. Descriptive comparisons do not account for all potential confounding variables, and statistical associations do not demonstrate causation.

The findings are also limited to the available dataset and the variables recorded within it. Additional clinical and operational information may be required to explain the observed patterns.

8. Conclusion

This project used Microsoft Excel to analyse emergency department performance, focusing on waiting times, patient satisfaction, arrivals and referral patterns.

The reported correlation coefficient of -0.0212 indicated a negligible linear relationship between waiting time and satisfaction among records with recorded scores. The chi-square test identified a statistically significant association between patient race and department referral, although Cramér’s V indicated that the association was very weak.

The findings provide a descriptive overview of the available records but do not establish causal relationships or demonstrate why the observed patterns occur. Further investigation of missing satisfaction data, clinical needs, staffing, referral definitions and other relevant factors would help inform evidence-based service improvement.